

# Computer Networks

## Data Link Control Protocols (Multiple Access Control Protocols) (Random Access Protocols: CSMA)

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# MAC protocols: taxonomy

Three broad classes:

- **Channel partitioning**

- Divide channel into smaller “pieces” (time slots, frequency, code)
- Allocate piece to node for exclusive use

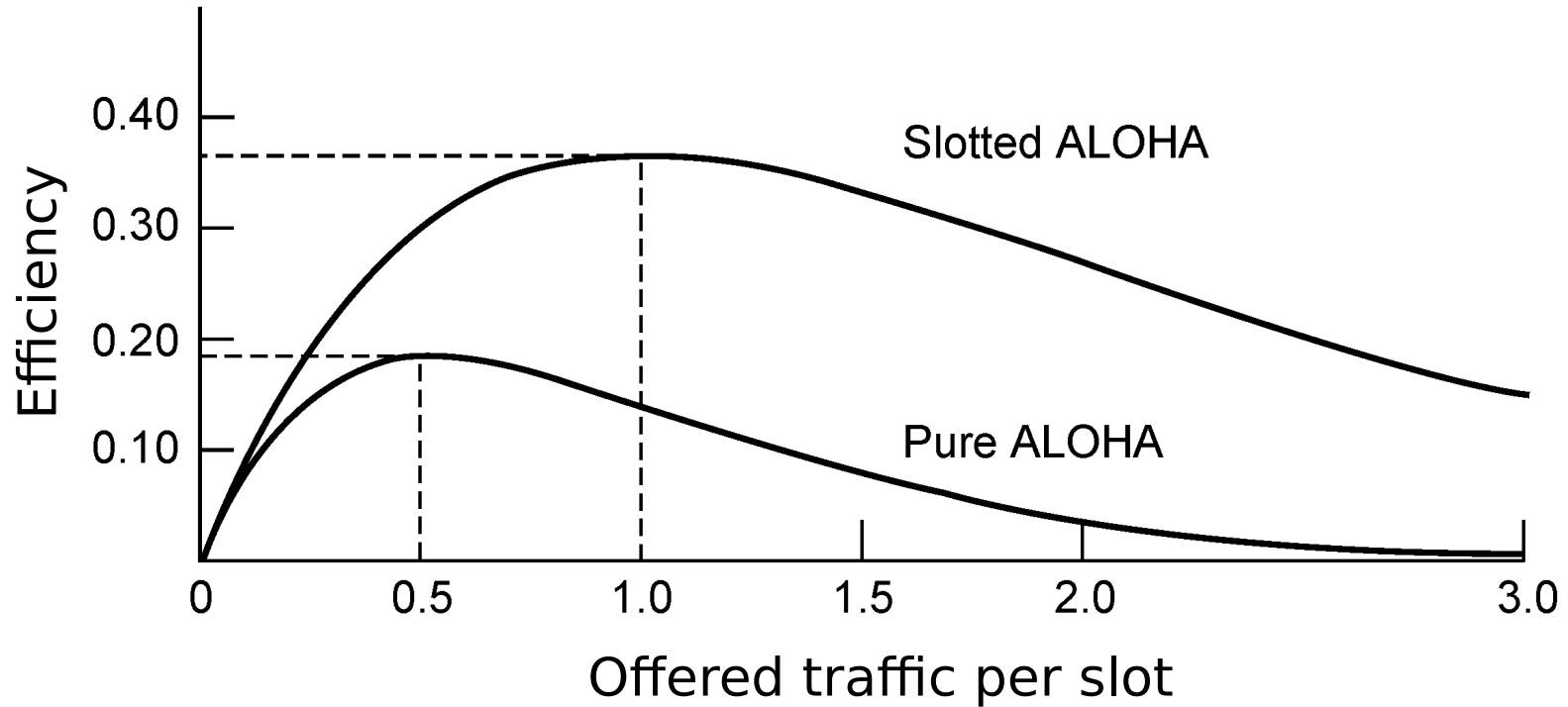
- **Taking turns”**

- Nodes take turns, but nodes with more to send can take longer turns

- **Random access**

- Use randomization for handling collisions
  - “Recover” from collisions
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# Pure ALOHA vs Slotted ALOHA



# CSMA (carrier sense multiple access)

Simple **CSMA**: listen before transmit

- If channel sensed **idle**: transmit entire frame
- If channel sensed **busy**: defer transmission
- Human analogy: don't interrupt others!



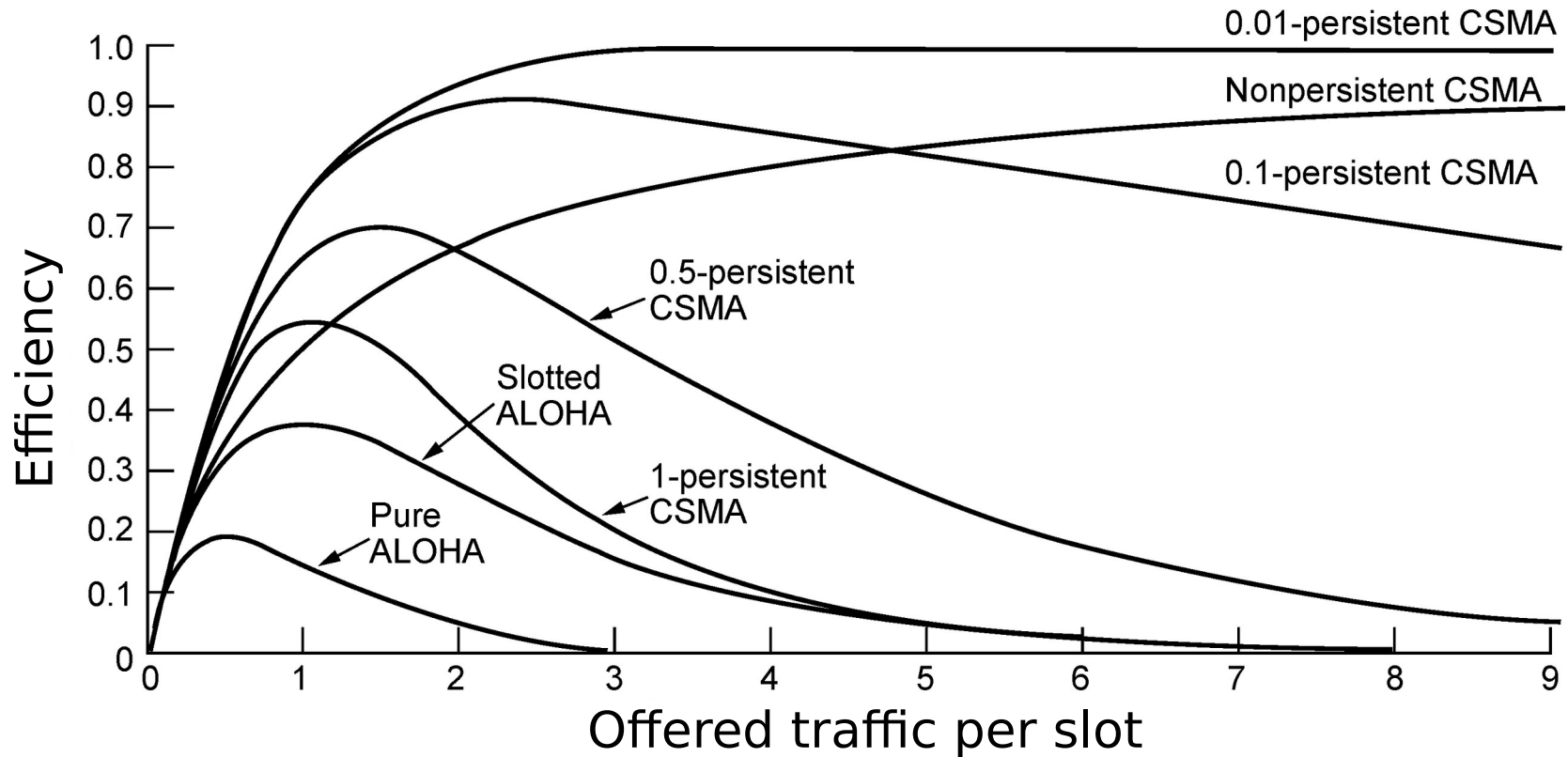
humans at a cocktail party  
(shared air, acoustical)

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# CSMA (carrier sense multiple access)

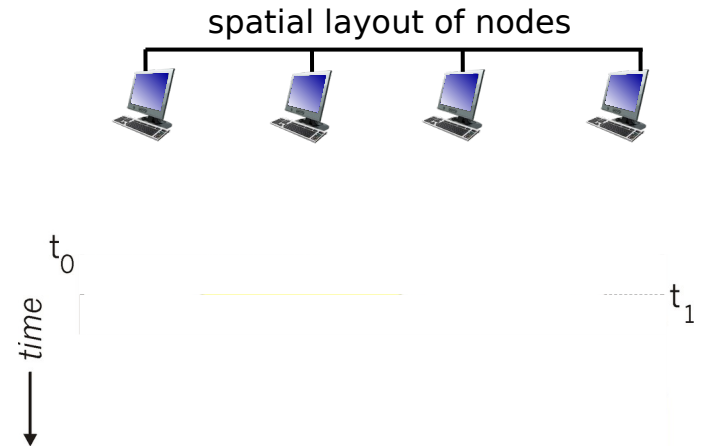
- 1-persistent CSMA:
    - If channel sensed idle: transmit with probability 1
    - If channel sensed busy: continuously sense → transmit immediately when channel becomes idle
    - If collision: wait for a random time, then start sensing
  - Nonpersistent CSMA:
    - If channel sensed busy: do not sense continuously → wait for a random amount of time and then sense again
    - Better channel utilization → longer delay than 1-persistent CSMA
  - p-persistent CSMA:
    - If channel sensed idle: transmit with probability  $p$
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# CSMA (carrier sense multiple access)



# CSMA: collisions

- Collisions can still occur with carrier sensing:
  - Propagation delay means two nodes may not hear each other's just-started transmission
  - Distance & propagation delay play role in determining collision probability
- **Collision:** entire packet transmission time wasted



# CSMA (carrier sense multiple access)

Simple **CSMA**: listen before transmit:

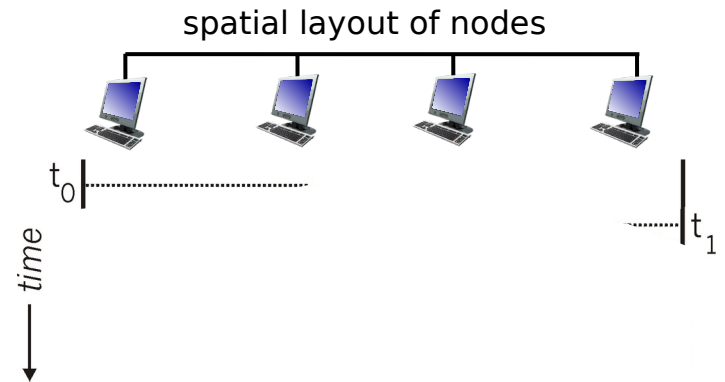
- If channel sensed idle: transmit entire frame
- If channel sensed busy: defer transmission
- Human analogy: don't interrupt others!

**CSMA/CD**: CSMA with collision detection

- Collisions detected within short time
  - Colliding transmissions aborted, reducing channel wastage
  - Collision detection easy in wired, difficult with wireless
  - Human analogy: the polite conversation
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# CSMA/CD

- CSMA/CD reduces the amount of time wasted in collisions
  - Transmission aborted on collision detection



# Ethernet CSMA/CD algorithm

1. Ethernet receives datagram from network layer, creates frame
  2. If Ethernet senses channel:
    - If **idle**: start frame transmission.
    - If **busy**: wait until channel idle, then transmit
  3. If entire frame transmitted without collision - done!
  4. If another transmission detected while sending: abort, send jam signal
  5. After aborting, enter **binary (exponential) backoff**:
    - After m-th collision, chooses K at random from  $\{0, 1, 2, \dots, 2^m - 1\}$
    - Wait for K slots, return step 2
    - More collisions: longer backoff interval
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# Ethernet: unreliable, connectionless

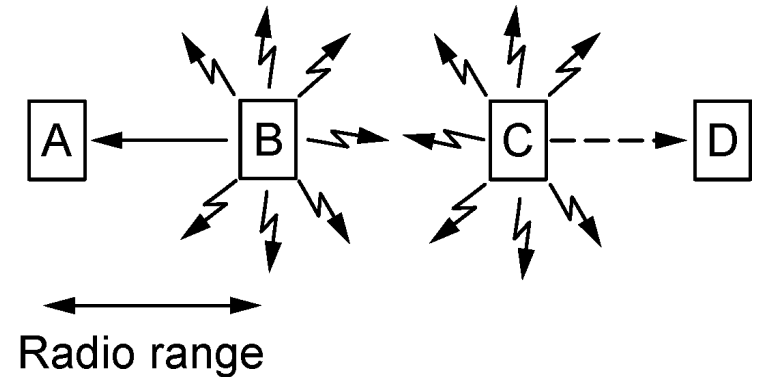
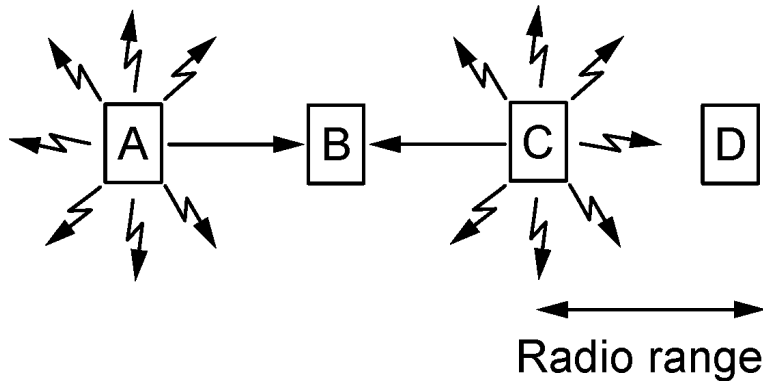
- **Connectionless:** No handshaking between sending and receiving NICs
  - **Unreliable:** Receiving NIC doesn't send ACKs or NAKs to sending NIC
    - Data in dropped frames may be recovered by the higher layer protocols (e.g., TCP), otherwise dropped data lost
  - Ethernet's MAC protocol: **1-persistent CSMA/CD with binary backoff**
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# Wireless $\neq$ CSMA/CD

- Transmitter cannot send/listen concurrently

# Wireless $\neq$ CSMA/CD

- Nodes have different coverage areas
  - Hidden terminal problem and Exposed terminal problem



# CSMA/CA: MACA → IEEE 802.11

- MACA: Sender and receiver uses a short handshake before transmitting
    - IEEE 802.11 uses a refinement of MACA
  - 802.11: CSMA - sense before transmitting
  - 802.11: no collision detection!
    - Goal: **avoid collisions**: CSMA/CollisionAvoidance
    - Hidden and exposed station problems are alleviated using RTS/CTS exchanges
    - In 802.11 RTS/CTS is optional
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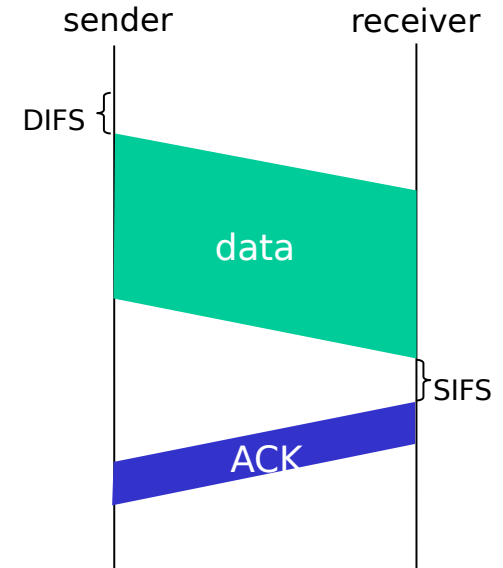
# CSMA/CA: IEEE 802.11

## 802.11 sender

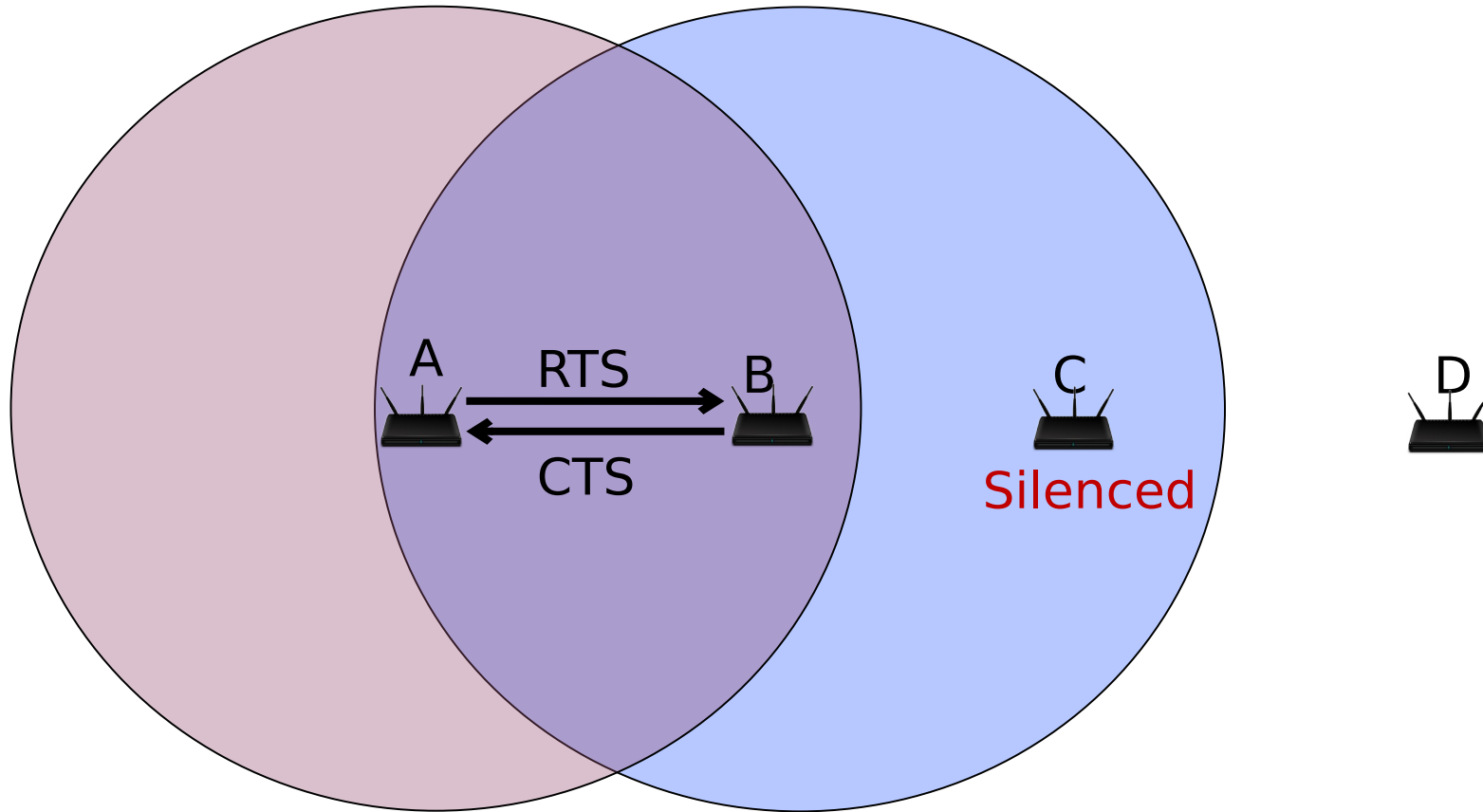
1. If sense channel idle for **DIFS** then  
transmit entire frame (no CD)
2. If sense channel busy then  
start random backoff time in range  $(0, w-1)$ ,  $w = CW_{\min}$   
timer counts down while channel idle  
transmit when timer expires  
if no ACK, increase random backoff interval  
(double  $w$  upto  $CW_{\max}$ ), repeat 2

## 802.11 receiver

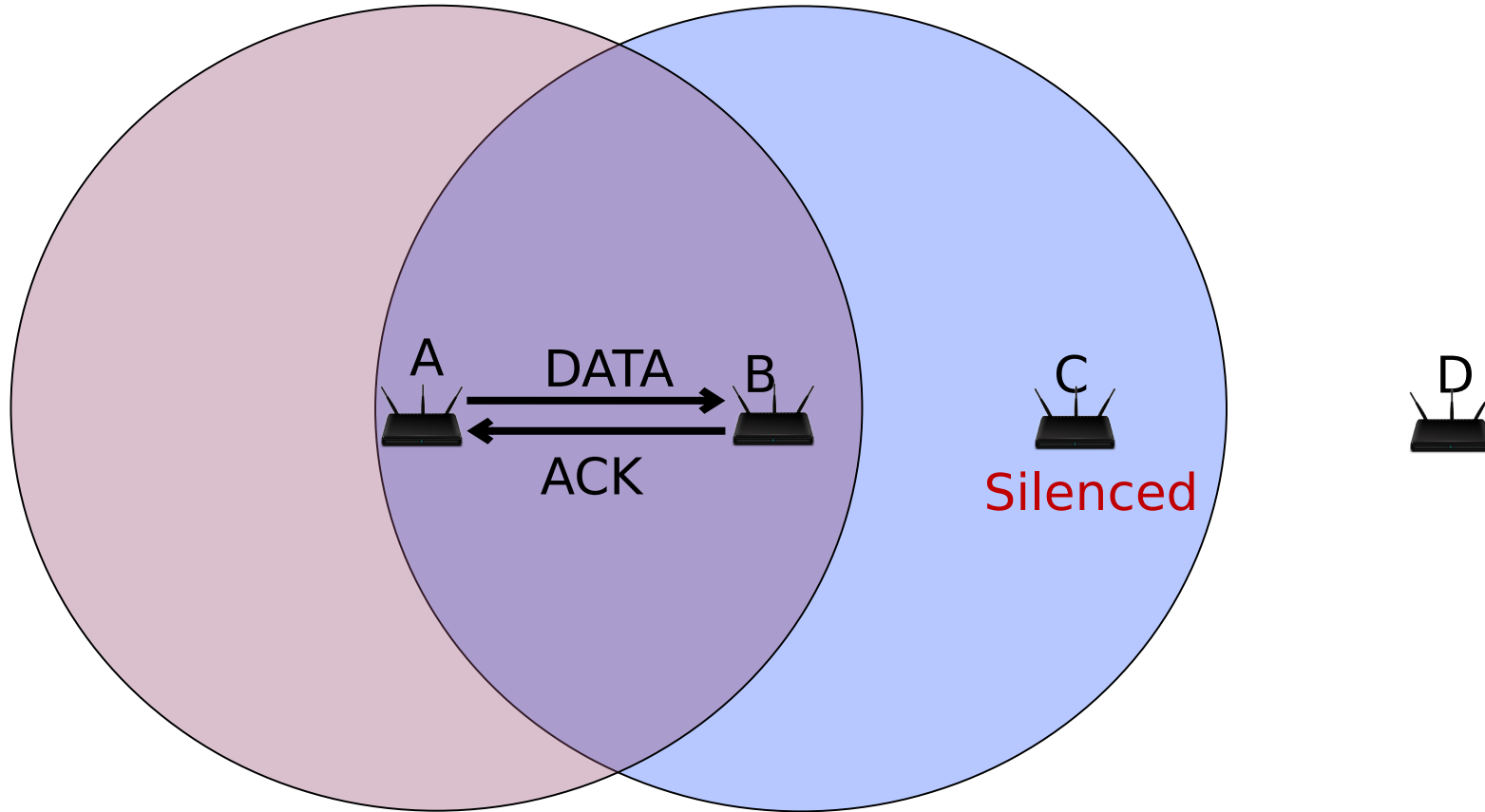
- If frame received OK  
return ACK after **SIFS** (ACK needed due to  
hidden terminal problem)



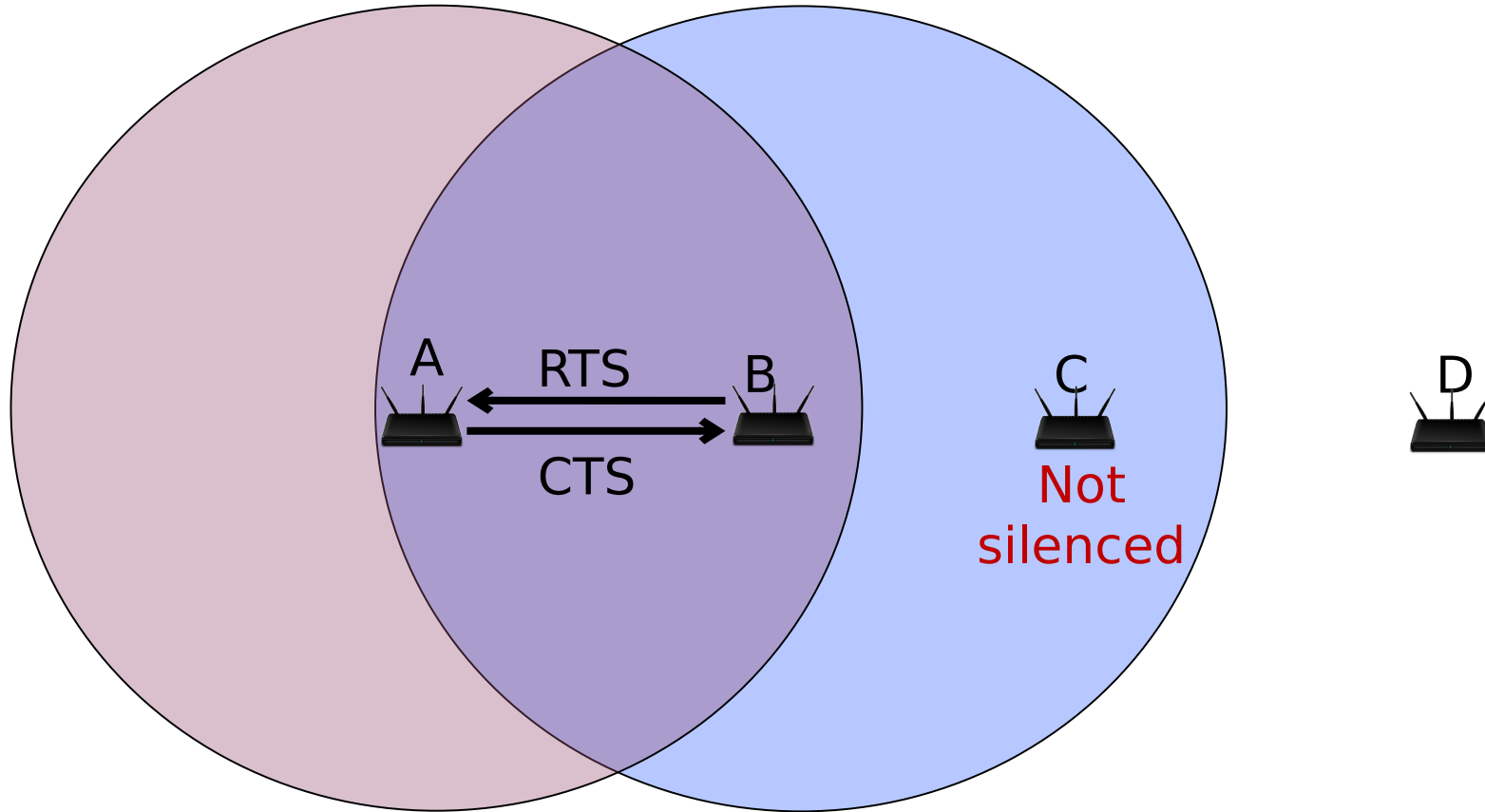
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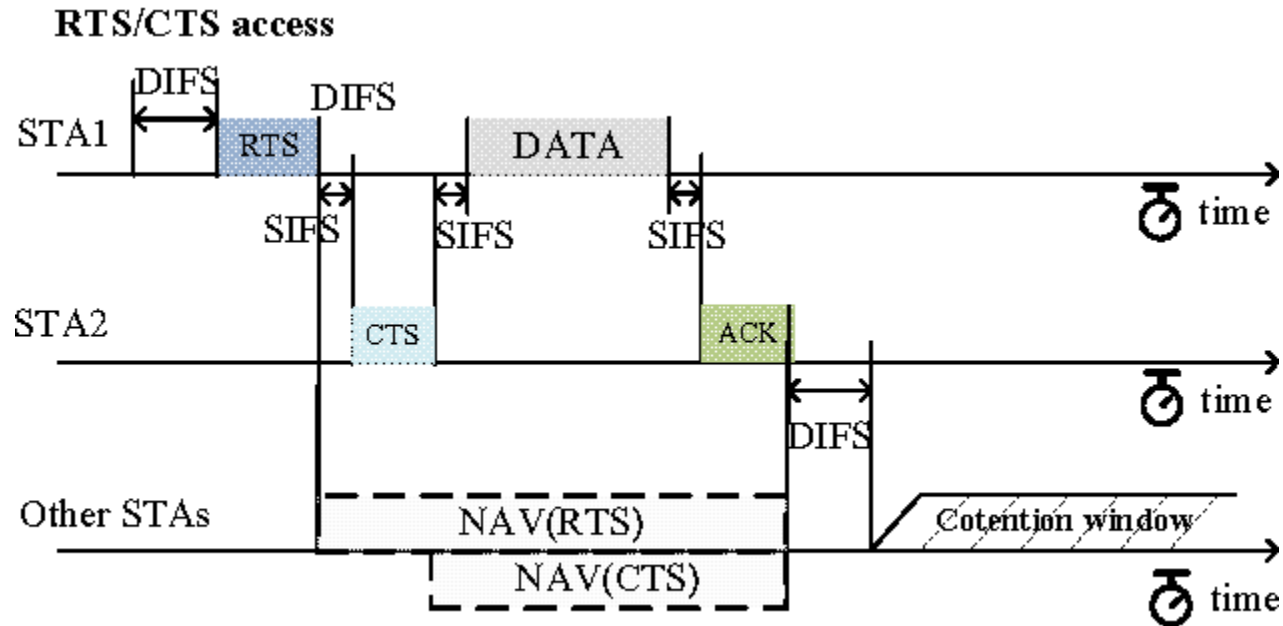
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# CSMA/CA: IEEE 802.11

- Sender first transmits small request-to-send (RTS) packet to the receiver
  - Receiver replies with clear-to-send CTS in response to RTS
  - Sender transmits data frame
  - Nodes receiving CTS defer transmissions
  - Nodes receiving RTS → defer one CTS time
  - Nodes receiving RTS but not CTS, free to send
  - RTS/CTS may still collide, but less likely (as they're short)
    - RTS/CTS exchange does not [entirely](#) solve the hidden and exposed station problem
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# CSMA/CA: IEEE 802.11



Src: [https://www.researchgate.net/publication/26429569\\_Autonomous\\_Power\\_Control\\_MAC\\_Protocol\\_for\\_Mobile\\_Ad\\_Hoc\\_Networks/figures?lo=1](https://www.researchgate.net/publication/26429569_Autonomous_Power_Control_MAC_Protocol_for_Mobile_Ad_Hoc_Networks/figures?lo=1)

# IEEE 802.11 Wireless LAN

| Standard | Frequency                 | Maximum Speed | Backwards compatibility |
|----------|---------------------------|---------------|-------------------------|
| 802.11   | 2.4 GHz                   | 2 Mbps        | -                       |
| 802.11a  | 5 GHz                     | 54 Mbps       | -                       |
| 802.11b  | 2.4 GHz                   | 11 Mbps       | -                       |
| 802.11g  | 2.4 GHz                   | 54 Mbps       | 802.11b                 |
| 802.11n  | 2.4 and 5 GHz             | 600 Mbps      | 802.11a/b/g             |
| 802.11ac | 5 GHz                     | 1300 Mbps     | 802.11a/n               |
| 802.11ad | 2.4 GHz, 5 GHz and 60 GHz | 7 Gbps        | 802.11a/b/g/n/ac        |

Src: <https://networkustad.com/2019/11/16/ieee-802-11-standards/>

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# MAC protocols: Comparison

Three broad classes:

- **Channel partitioning**
    - Share channel efficiently and fairly at high load
    - Inefficient at low load: delay in channel access,  $1/N$  bandwidth allocated even if only 1 active node!
  - **Taking turns**
    - Look for best of both worlds
    - Single point of failure
  - **Random access**
    - Efficient at low load: single node can fully utilize channel
    - High load: collision overhead
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# Summary

## □ Random access based MAC protocols:

- CSMA
  - CSMA/CD
  - CSMA/CA
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- Comparison of the MAC protocols
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